CS476: Natural Language Processing

Course Project Report

Arabic Legal Document Classification

Using the ALARB Dataset — Saudi Commercial Court Cases

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Task Distribution

Task
Phase 1: EDA & Dataset Loading
Phase 2: Label Engineering (Task A & B)
Phase 3: Arabic Preprocessing Pipeline
Phase 4: Feature Extraction
Phase 5: Model Training (LR, SVM, ANN, BiLSTM, Task B)
Phase 6: Evaluation, Experiments, Error Analysis
Phase 7: Live Demo (Gradio)

Table Of Contents

1. Introduction **4**
2. Dataset **5**
2.1 About the ALARB Dataset 5
2.2 Dataset Statistics 5
2.3 Task A — Verdict Class Distribution (Training Set) 5
2.4 Task B — Topic Class Distribution (Training Set) 6
2.5 Filtering Decisions 6
3. Classification Task **7**
3.1 Task A — Verdict Prediction 7
3.2 Task B — Legal Topic Classification 7
3.3 Why These Two Tasks 7
4. Data Preprocessing **8**
4.1 Alef, Taa Marbuta, and Hamza Normalization 8
4.2 Diacritics and Tatweel Removal 8
4.3 Legal-Domain Token Handling 8
4.4 Arabic Stopword Removal 8
4.5 Stemming and Lemmatization 8
4.6 Train/Test Split 8
5. Feature Extraction **10**
5.1 Bag of Words 10
5.2 TF-IDF 10
5.3 N-grams 10
5.4 Word Embeddings 10
6. Model Selection and Training **12**
6.1 Logistic Regression 12
6.2 Linear SVM 12
6.3 Feedforward ANN 12
6.4 BiLSTM 12
6.5 Task B Models 12
7. Evaluation **14**
7.1 Metrics 14
7.2 Task A — Verdict Prediction Results 14
7.3 Task B — Legal Topic Classification Results 14
8. Experiment Results and Analysis **15**
Scenario 1 — BoW vs TF-IDF 15
Scenario 2 — Unigram vs Bigram 15
Scenario 3 — Word N-grams vs Character N-grams 15
Scenario 4 — Traditional ML vs Deep Learning 16
Scenario 5 — Sparse TF-IDF vs Dense Embeddings 16
9. Error Analysis **17**
9.1 Overall Error Rate 17
9.2 Pattern 1 — Rejected vs Accepted Confusion (268 cases) 17

9.3 Pattern 2 — Minority Class Errors (129 cases) 17
9.4 Pattern 3 — Settlement Confusion (52 cases) 17
9.5 Pattern 4 — Document Length 18
10. Ethical Impact Assessment 19
10.1 Bias in Legal Predictions 19
10.2 Risk of Over-Reliance 19
10.3 Fairness Across Case Types 19
10.4 Privacy of Litigants 19
10.5 Responsible Deployment Guidelines 20
11. Conclusion 21
12. References 22

1. Introduction

Saudi commercial courts handle thousands of cases every year, and going through all these records manually takes a lot of time and effort. In this project, we built a system that reads Arabic court case documents and automatically classifies them — either by predicting the verdict outcome or by identifying what legal topic the case is about.

We used the ALARB dataset, which contains 13,028 real Saudi commercial court cases. Our system handles two tasks: Task A predicts the verdict (for example, whether the claim was accepted, rejected, or settled), and Task B identifies the legal subject area of the case (for example, sales contracts or construction disputes).

Working with Arabic legal text is not straightforward. Legal Arabic uses very formal language that is quite different from everyday speech. Words are written differently depending on context, and the same legal concept can appear in many different forms due to Arabic morphology. On top of that, not many labeled Arabic legal datasets exist, which makes this kind of research challenging and important at the same time.

This report covers everything we did: how we cleaned and prepared the data, how we turned text into numbers the model can understand, which models we trained and why, and what results we got. We also discuss where the system makes mistakes and what ethical concerns need to be considered before using a tool like this in real legal settings.

2. Dataset

2.1 About the ALARB Dataset

The ALARB (Arabic Legal Argument Reasoning Benchmark) dataset was published by Abu Shairah et al. in 2025 and is available on Hugging Face at THIQAH-RD/ALARB. It contains structured Saudi commercial court cases with four fields per case:

- case_facts — the claims of the plaintiff and the response of the defendant. This is the only field we used as model input.
- court_reasoning — the step-by-step reasoning written by the court. We did not use this as input because it already contains the answer (data leakage).
- verdict — the court's final decision in free text. We used this as the label source for Task A after cleaning and normalizing it.
- applicable_laws — a list of legal articles cited in the case. We parsed this offline to create topic labels for Task B.

2.2 Dataset Statistics

Item	Value
Total cases (raw)	13,028
Training set (80%)	10,014 cases
Test set (20%)	2,504 cases
Task A classes	5 verdict classes
Task B training cases	9,291 (after excluding أخرى)
Task B test cases	2,333
Task B classes	5 topic classes
Raw vocabulary size	11,848 unique tokens
BoW vocabulary	11,042 tokens
TF-IDF Uni+Bigram vocabulary	230,353 tokens

2.3 Task A — Verdict Class Distribution (Training Set)

Verdict Class	Arabic Label	Count	Percentage
Fully Accepted	مقبولة كلياً	5,976	59.7%
Rejected	مرفوضة	1,901	19.0%
Settlement	صلح	961	9.6%
Jurisdictional Dismissal	عدم اختصاص	701	7.0%
Lapsed	انقضاء	475	4.7%

2.4 Task B — Topic Class Distribution (Training Set)

Topic Class	Arabic Label	Count
Commercial Procedures	إجراءات تجارية	4,091
Bankruptcy & Enforcement	إفلاس وتنفيذ	2,081
Litigation Procedures	مرافعات	1,610
Evidence & Proof	إثبات	1,167
Companies & Other	شركات وأخرى	342

2.5 Filtering Decisions

We removed the مقبولة جزئياً (partially accepted) class from Task A because its language overlaps heavily with both full acceptance and rejection, making reliable classification very difficult. Cases with no parseable law citation (labeled أخرى) were removed from Task B training and evaluation. The class imbalance in Task A — where مقبولة كلياً makes up nearly 60% of the data — was handled in all models using class_weight='balanced'.

3. Classification Task

3.1 Task A — Verdict Prediction

Given the case_facts field, the model predicts one of five verdict classes. This is a single-label five-class classification problem. The input is only what was known before the court made its decision (the plaintiff and defendant statements), not the court's reasoning or the verdict text itself.

Class	Arabic	Meaning
Fully Accepted	مقبولة كلياً	Court ruled entirely in favor of the plaintiff
Rejected	مرفوضة	Court rejected all plaintiff claims
Settlement	صلح	Parties reached a mutual agreement during proceedings
Jurisdictional Dismissal	عدم اختصاص	Case transferred or dismissed on jurisdiction grounds
Lapsed	انقضاء	Case closed due to withdrawal or procedural lapse

3.2 Task B — Legal Topic Classification

Given the same case_facts field, the model predicts the legal subject area of the dispute. Labels were created by parsing the applicable_laws column offline and mapping the most frequently cited laws to topic categories. This column was never fed into the model — it was only used to generate labels during data preparation.

3.3 Why These Two Tasks

We chose Task A because verdict prediction is the most practically useful task — it directly helps lawyers and judges quickly assess likely outcomes for new cases. We chose Task B as our second task because it tests a different aspect of the same documents: understanding what the case is about rather than how it was decided. Together, the two tasks give a fuller picture of what the NLP pipeline can do with legal case descriptions.

4. Data Preprocessing

All preprocessing was implemented in Python using a single preprocess_text() function that supports two modes: stemming (used for BoW, TF-IDF, and n-gram features) and lemmatization (used for word embeddings). We implemented the following five Arabic-specific techniques:

4.1 Alef, Taa Marbuta, and Hamza Normalization

Arabic has several spelling variants of the same letter. For example, the word for 'court' can be written as المحكمة or المحكمه depending on the writer. We normalized all Alef forms (أ, إ, آ) to bare Alef (ا), Taa Marbuta (ة) to Haa (ه), and Hamza variants to their base forms. Without this step, the same word would appear as multiple different tokens in the vocabulary.

4.2 Diacritics and Tatweel Removal

Legal documents are rarely vowelized, but some case files contain diacritical marks (tashkeel) that add noise. We removed all diacritics using the full Unicode diacritics block. We also removed the tatweel character (-) which is used for visual stretching and carries no meaning.

4.3 Legal-Domain Token Handling

Rather than removing numbers and dates, we replaced them with meaningful placeholder tokens. Hijri and Gregorian dates became DATE_TOKEN, monetary amounts became MONEY_TOKEN, case reference numbers became CASE_TOKEN, and other numbers became NUM_TOKEN. This preserves the information that a date or amount was mentioned without letting specific values dominate the vocabulary.

4.4 Arabic Stopword Removal

We used a custom Arabic stopword list built specifically for legal text. The most important difference from general stopword lists is that we kept negation particles like لم ,لا, and ليس. In legal Arabic, these particles completely change the meaning of a sentence — the difference between مقبولة (accepted) and لا مقبولة (not accepted) is exactly these words. Removing them would destroy critical information for verdict prediction.

4.5 Stemming and Lemmatization

We applied ISRI stemming (via NLTK) for Task A models because it reduces vocabulary size efficiently — from 11,848 raw tokens down to a more manageable set — without needing external dependencies. For word embeddings, we used Farasa lemmatization instead, because it preserves more of the original word meaning, which matters for semantic similarity in the embedding space. We ran both on a sample of 50 legal terms and compared the output to confirm that stemming did not incorrectly collapse distinct legal concepts.

4.6 Train/Test Split

We split the data 80/20 into training (10,014 cases) and test (2,504 cases) using stratified sampling so that each verdict class appears in the same proportion in both sets. We used random_state=42 throughout for reproducibility.

5. Feature Extraction

5.1 Bag of Words

Our first feature method was a simple CountVectorizer fitted on the stemmed training text. It produced a sparse matrix of shape (10,014 x 11,042) — meaning 10,014 training documents each represented by 11,042 word frequency counts. BoW treats each document as an unordered bag of words, so it captures which legal terms appear but not their order or relative importance. We used it as a baseline to compare against more sophisticated methods.

5.2 TF-IDF

TF-IDF improves on BoW by giving higher weight to terms that are specific to certain documents and lower weight to terms that appear in almost every case. We ran three configurations. Unigram TF-IDF produced an 11,042-feature matrix. Adding bigrams grew the vocabulary to 230,353 features. Capping at 20,000 features gave a manageable matrix while keeping the most important bigrams. The unigram+bigram configuration consistently gave the best results in our experiments, so we used it as the main feature set for LR and SVM training.

5.3 N-grams

We tested four n-gram configurations: word bigrams (10,014 x vocabulary), word trigrams, character bigrams, and character trigrams. Word bigrams capture legal two-word phrases like رفض الدعوى (rejection of claim) that are more specific than individual words. Character trigrams are more robust to Arabic morphology because they match sub-word patterns, but they produce a much larger feature space. In our experiment scenarios, word bigrams performed comparably to character trigrams while being computationally cheaper.

5.4 Word Embeddings

We trained a Word2Vec Skip-gram model directly on the ALARB training corpus using lemmatized text (vector_size=300, window=5, min_count=5, epochs=3). The model learned 5,574 Arabic legal word vectors. We initially planned to use the pre-trained Arabic FastText model (cc.ar.300.bin) but it requires 4.2 GB of RAM to load, which exceeded the Google Colab free tier limits. Training on the ALARB corpus turned out to be a better choice anyway — the resulting embeddings are tuned specifically to Saudi commercial court vocabulary. However, the OOV rate was 52.95%, meaning more than half of unique tokens in the test set were not seen during Word2Vec training. This explains why embedding-based models underperformed TF-IDF in our experiments.

Feature Method	Train Shape	Vocab/Dim	Input Text
Bag of Words	(10,014 × 11,042)	11,042 tokens	case_facts_stem
TF-IDF Unigram	(10,014 × 11,042)	11,042 tokens	case_facts_stem
TF-IDF Uni+Bigram	(10,014 × 230,353)	230,353 tokens	case_facts_stem
TF-IDF Uni+Bi (20k)	(10,014 × 20,000)	20,000 tokens	case_facts_stem

Feature Method	Train Shape	Vocab/Dim	Input Text
Word Bigrams	(10,014 × vocab)	word bigrams	case_facts_stem
Char Trigrams	(10,014 × vocab)	char trigrams	case_facts_stem
Word2Vec Embeddings	(10,014 × 300)	300-dim dense	case_facts_lemma

6. Model Selection and Training

6.1 Logistic Regression

Logistic Regression with L2 regularization was our first traditional ML model. It works well for high-dimensional sparse text features because it learns a weight for each vocabulary term indicating how strongly that term points toward each verdict class. We used GridSearchCV with 3-fold stratified cross-validation, optimizing macro F1. The best configuration was C=10, class_weight='balanced', achieving a cross-validation macro F1 of 0.7193 on TF-IDF Uni+Bigram features.

6.2 Linear SVM

Linear SVM was our second traditional ML model and ended up being the best overall performer. SVMs find the widest possible margin between classes in the feature space, which makes them particularly good at handling the high-dimensional TF-IDF matrices we used. GridSearchCV found C=1, class_weight='balanced' as the best configuration, with a cross-validation macro F1 of 0.7196 — slightly better than Logistic Regression.

6.3 Feedforward ANN

We built a feedforward Artificial Neural Network using Keras with two dense hidden layers (256 units then 128 units, both with ReLU activation), dropout regularization (0.3 and 0.2), and a softmax output layer. The model was trained on TF-IDF 20k features with Adam optimizer and EarlyStopping (patience=3). The ANN achieved an accuracy of 0.7316 and macro F1 of 0.6896 — lower than both linear models. This is likely because the dataset size (10,014 cases) is not large enough for the ANN to learn better non-linear representations than what TF-IDF + linear models already capture.

6.4 BiLSTM

The BiLSTM was our deep learning model for handling sequential text. The architecture uses a learned embedding layer (30,000 token vocabulary, 128-dim embeddings), SpatialDropout1D (0.3), a Bidirectional LSTM (128 units per direction), GlobalMaxPooling1D, a dense layer (64 units), and softmax output. We used GlobalMaxPooling1D specifically because it picks the most informative hidden state across the whole sequence — this works better than taking only the final hidden state for long legal documents with up to 300 tokens. The BiLSTM achieved accuracy 0.7596 and macro F1 0.7064, outperforming the ANN but still behind the linear models.

6.5 Task B Models

For topic classification we trained LR and SVM on TF-IDF Uni+Bigram features (max 20,000) with topic_label as the target. Training size was 9,291 cases (cases labeled أخرى were excluded). GridSearchCV found C=10 for LR (CV macro F1: 0.5338) and C=0.1 for SVM (CV macro F1: 0.5320). The lower cross-validation scores compared to Task A reflect the greater difficulty of topic classification.

Model	Best Params	CV Macro F1	Task
Logistic Regression	C=10, balanced	0.7193	A
Linear SVM	C=1, balanced	0.7196	A
ANN	EarlyStopping p=3	—	A
BiLSTM	EarlyStopping p=3	—	A
LR (Topic)	C=10, balanced	0.5338	B
SVM (Topic)	C=0.1, balanced	0.5320	B

7. Evaluation

7.1 Metrics

We evaluated all models on the held-out test set (2,504 cases) using five metrics: Accuracy, Weighted Precision, Weighted Recall, Macro F1, and Weighted F1. We used Macro F1 as the primary ranking metric because it treats all five classes equally regardless of how many examples each class has — this matters here because مقبولة كلياً has 5,976 training examples while انقضاء has only 475. We also plotted confusion matrices for all models to understand which classes were being confused with each other.

7.2 Task A — Verdict Prediction Results

Model	Features	Accuracy	Macro F1	Weighted F1
Linear SVM	TF-IDF Uni+Bigram	0.8003	0.7225	0.7951
Logistic Regression	TF-IDF Uni+Bigram	0.7927	0.7205	0.7915
BiLSTM	Padded sequences	0.7596	0.7064	0.7635
ANN	TF-IDF 20k	0.7316	0.6896	0.7453

7.3 Task B — Legal Topic Classification Results

Model	Features	Accuracy	Macro F1	Weighted F1
Task B Linear SVM	TF-IDF Uni+Bigram 20k	0.5765	0.5501	0.5799
Task B Logistic Regression	TF-IDF Uni+Bigram 20k	0.5744	0.5490	0.5774

Task B results are noticeably lower than Task A. This is expected — predicting what a case is about is harder than predicting the verdict, because different legal topics often use very similar Arabic vocabulary. A sales contract dispute and a construction contract dispute both involve payment terms, amounts, and delivery obligations, making them hard to tell apart from the case facts alone.

8. Experiment Results and Analysis

We ran 14 experiment scenarios by combining 2 models (LR and SVM) with 7 feature sets. Here are the full results sorted by Macro F1:

Scenario	Accuracy	Macro F1	Weighted F1
Linear SVM + TF-IDF Uni+Bi	0.8003	0.7225	0.7951
Logistic Regression + TF-IDF Uni+Bi	0.7927	0.7205	0.7915
Logistic Regression + TF-IDF 20k	0.7815	0.7172	0.7837
Linear SVM + TF-IDF 20k	0.7823	0.7113	0.7809
Linear SVM + Word Bigrams	0.7891	0.7049	0.7836
Logistic Regression + Word Bigrams	0.7843	0.7030	0.7811
Linear SVM + TF-IDF Unigram	0.7384	0.6479	0.7400
Logistic Regression + TF-IDF Unigram	0.7105	0.6338	0.7194
Logistic Regression + Char Trigrams	0.6901	0.6027	0.6960
Logistic Regression + BoW	0.6913	0.5965	0.6965
Linear SVM + Word2Vec Embeddings	0.6993	0.5955	0.7044
Linear SVM + BoW	0.6961	0.5937	0.6991
Linear SVM + Char Trigrams	0.6885	0.5878	0.6906
Logistic Regression + Word2Vec	0.6110	0.5388	0.6370

Scenario 1 — BoW vs TF-IDF

Switching from BoW (Macro F1: 0.5965 for LR) to TF-IDF Uni+Bigram (Macro F1: 0.7205 for LR) gave a 12.4 percentage point improvement. BoW counts every word the same way, so common legal phrases like بسم الله الرحمن الرحيم that appear in every case get the same weight as verdict-specific terms. TF-IDF fixes this by lowering the weight of words that appear in almost all documents.

Scenario 2 — Unigram vs Bigram

Adding bigrams to TF-IDF improved Macro F1 from 0.6479 (SVM unigram) to 0.7225 (SVM uni+bigram) — a 7.5 point gain. This makes sense for legal Arabic because many verdict signals are two-word phrases: رفض الدعوى means rejection of the claim, while رفض alone could appear in many other contexts.

Scenario 3 — Word N-grams vs Character N-grams

Word bigrams (SVM Macro F1: 0.7049) outperformed character trigrams (SVM Macro F1: 0.5878) despite character n-grams being theoretically more robust to Arabic morphology. We think this is because our stemming already handled morphological variation, so character-level features added noise rather than useful signal in this case.

Scenario 4 — Traditional ML vs Deep Learning

Linear SVM (Macro F1: 0.7225) outperformed both BiLSTM (0.7064) and ANN (0.6896). This surprised us at first, but it makes sense: the strong discriminative vocabulary in legal Arabic (specific terms that directly indicate verdicts) is well captured by linear TF-IDF weights. Deep learning needs more data to learn useful representations from scratch.

Scenario 5 — Sparse TF-IDF vs Dense Embeddings

TF-IDF consistently outperformed Word2Vec embeddings. The best embedding scenario (SVM + Word2Vec Macro F1: 0.5955) was 12.7 points below the best TF-IDF scenario. The main reason is the high OOV rate of our Word2Vec model: 52.95% of test tokens were not in the training vocabulary, so those words got zero vectors. This lost a lot of information. Future work using AraBERT or AraLegal-BERT — which handle OOV through subword tokenization — would likely close this gap significantly.

9. Error Analysis

9.1 Overall Error Rate

The best model (Linear SVM + TF-IDF Uni+Bigram) misclassified 519 out of 2,504 test cases — an error rate of 20.7%. Here are the top 10 most common confusion pairs:

True Label	Predicted As	Count
مرفوضة	مقبولة كلياً	134
مقبولة كلياً	مرفوضة	134
عدم اختصاص	مرفوضة	42
عدم اختصاص	مقبولة كلياً	37
مرفوضة	عدم اختصاص	30
انقضاء	مقبولة كلياً	24
صلح	مقبولة كلياً	21
مقبولة كلياً	صلح	18
مقبولة كلياً	عدم اختصاص	15
صلح	انقضاء	13

9.2 Pattern 1 — Rejected vs Accepted Confusion (268 cases)

The most frequent error is confusing مرفوضة (rejected) with مقبولة كلياً (accepted) in both directions — 134 cases each way. Both classes use the word إلزام (obligation). In accepted cases it appears as إلزام المدعى عليه بالسداد (obligating the defendant to pay), while in rejected cases it appears near negation particles: لا يلزم (not obligated). The model sometimes misses the negation because لا is a very common word that appears in many contexts.

9.3 Pattern 2 — Minority Class Errors (129 cases)

عدم اختصاص and انقضاء together account for 129 misclassifications despite making up only 11.7% of the training data. Even with class_weight='balanced', these classes have fewer training examples, so the model is less confident about their distinctive patterns. Many عدم اختصاص cases get predicted as مرفوضة because both involve the plaintiff not getting what they wanted, and the surface language is similar.

9.4 Pattern 3 — Settlement Confusion (52 cases)

Settlement verdicts (صلح) are confused with مقبولة كلياً in 21 cases and with انقضاء in 13 cases. Settlement cases contain obligation language (إلزام) because the court enforces the agreement, which looks like an accepted claim. The distinctive words for settlement — اتفاق الطرفين (mutual agreement) and التراضي (amicable resolution) — are sometimes absent or very short in the case facts, giving the model less to work with.

9.5 Pattern 4 — Document Length

When we looked at the word counts of misclassified cases, we found that they tend to be longer than correctly classified cases on average. Longer documents contain more boilerplate legal language, which dilutes the specific terms that signal the verdict. This suggests that a model capable of focusing on the most relevant parts of a document — such as an attention-based transformer — would likely reduce errors on longer cases.

10. Ethical Impact Assessment

10.1 Bias in Legal Predictions

Our system learned from historical court decisions. If the past decisions contained systematic biases — for example, if certain types of plaintiffs consistently received less favorable verdicts — the model will learn to replicate those patterns. We noticed in our results that the model performs much better on the majority class مقبولة كلياً (accuracy much higher than for minority classes like انقضاء). This means the system gives less reliable predictions for the less common case types, and litigants whose cases fall into those categories would be worse served by this tool.

Another concern is that our Task B labels were created by a rule-based mapping from cited laws to topic categories. This mapping was designed by our team and may not reflect how legal experts would categorize cases. If lawyers or judges used our topic labels to route cases, errors in our label engineering could send cases to the wrong legal category.

10.2 Risk of Over-Reliance

Our best model is wrong about 1 in 5 times (20.7% error rate). In most domains, 80% accuracy would be considered quite good. In legal settings, a wrong classification can affect someone's rights, money, and life. If a judge or legal aid organization relied too heavily on our system without checking its outputs, those errors could cause real harm. We strongly believe this tool should only be used to help humans make decisions — not to make decisions on its own.

There is also a risk that the existence of an automated system creates pressure to move faster through cases, reducing the time available for careful human judgment. Courts and legal organizations deploying such tools should establish clear policies that automated predictions are always reviewed by qualified humans.

10.3 Fairness Across Case Types

Our experiment results show that performance is uneven across verdict classes. The majority class مقبولة كلياً is classified very well, while minority classes like انقضاء (only 475 training examples) have much higher error rates. If this tool were deployed to assist with case triage, plaintiffs with cases in minority categories would systematically receive less accurate assistance. Regular fairness audits across all case types should be conducted before and during any real-world deployment.

10.4 Privacy of Litigants

The ALARB dataset contains real Saudi commercial court records. Even though the dataset is publicly available for research, the cases involve real people and businesses whose financial disputes are documented in our training data. Any production system built on this technology should anonymize litigant information before processing, comply with Saudi Arabia's Personal Data Protection Law (PDPL), and ensure that it is not possible to identify individual litigants from model outputs or intermediate representations. We did not collect

any new data in this project, but future work that expands the dataset would need to address these privacy concerns carefully.

10.5 Responsible Deployment Guidelines

Based on our experience building this system, we propose the following guidelines for responsible deployment:

1. Human review is mandatory: Every automated classification must be reviewed by a qualified legal professional before any action is taken based on it.
2. Confidence scores must be shown: The system already outputs confidence percentages for each class. Low-confidence predictions should trigger additional scrutiny.
3. Scope must be clearly defined: The system should only be used for verdict prediction and topic classification on commercial court cases. Using it for other legal tasks or other jurisdictions without retraining is not appropriate.
4. Regular performance audits: The system should be re-evaluated at least once a year to check whether its accuracy has degraded as legal language and law changes over time.
5. Affected parties must be informed: Any litigant whose case was processed by an automated system should be told that automated tools were used and should be able to request fully human review.
6. Data updates: As Saudi commercial law evolves, the training data and models must be updated to reflect current legal frameworks.

11. Conclusion

We built an end-to-end Arabic legal NLP pipeline that classifies Saudi commercial court cases by verdict outcome (Task A) and by legal topic (Task B). Our best model — Linear SVM with TF-IDF Unigram+Bigram features — achieved a Macro F1 of 0.7225 on Task A and our topic models achieved 0.5501 Macro F1 on Task B.

The most interesting finding from our experiments was that traditional linear models outperformed both deep learning models (BiLSTM and ANN) on this dataset. We believe this is because legal Arabic verdict prediction is largely vocabulary-driven — specific terms like رفض ,إلزام, and صلح are strong signals that TF-IDF captures well through its weighting scheme. Deep learning did not have enough training data to learn better representations.

We also found that the high OOV rate in our domain-trained Word2Vec model (52.95%) was a significant limitation. A pre-trained transformer model like AraLegal-BERT — which uses subword tokenization and was pre-trained on large Arabic legal corpora — would likely solve this problem and improve classification accuracy substantially. This is the most important direction for future work.

Other limitations include the exclusion of partially accepted cases (مقبولة جزئياً) and the reliance on rule-based label engineering for both tasks. A future version with human-annotated labels and more diverse training data would produce more reliable results. Despite these limitations, we believe this project demonstrates that NLP can meaningfully assist with Arabic legal document processing, and we hope it contributes to the growing body of Arabic legal NLP research.

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